



Routing

ROUTING
in
PHYSICAL DESIGN

INTEGRATED CIRCUIT COMPILER (ICC2)

Only for chip to startup (C2S) Training

GOALS OF ROUTING

Training

- The goal of the routing phase is to →
- Route all the signal nets with minimal physical DRC violations
 - Optimize data-path logic for timing, DRCs, and leakage power

Only for C++



Inputs to Routing

- CTS database (.def)
- Timing Library (.lib/.db)
- Physical Library (.lef)
- Tech Library (.tf)
- Cap Table (.itf/.tluplus)
- Constraints file (.sdc)

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Routing Flow

➤ Clock nets should have been routed already

➤ Signal nets are routed first – no cell optimization

route_auto

Routing Performs:

Global Routing → Track Assignment → Detail Routing

➤ Afterwards, post-route optimization is performed

route_opt

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GLOBAL ROUTING

➤ GR assigns nets to specific metal layers and global routing cells (Gcells)

➤ No metal traces are created

- Uses colored zero-width lines to indicate global routes and assigned metal layers

➤ GR tries to avoid congested Gcells while minimizing detours

- Congestion exists when more tracks are needed than available
- Detours increase wire length (Delay)

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ROUTE/TRACK ASSIGNMENT

➤ Track Assignments (TA)

- Assigns each net to a specific track within each global routing cell

➤ Attempts to

- Make long, straight traces
- Reduce the number of vias

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DETAILED ROUTING

- Creates the actual metal layer shapes for interconnection
- Addresses any Design Rule Check (DRC) violations that arise after track assignment.
- Uses small regions (SBoxes) of a fixed size to repair these violations.
- Progresses through the entire design, box by box, until the routing process is complete.